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INDIVIDUAL ASSIGNMENT

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1. Introduction

Marcotting, also known as air-layering, is a technique used in commercial agriculture to cultivate young sapling from grown trees, mainly fruit trees that have wooden stems. This method involves the growth of roots from the branch of the parent plant, forming a new plant that is ready to be planted.

2. Benefits

There are multiple reasons of using marcotting method instead of other planting methods such as direct seeding, transplanting and cutting, as stated below:

a. Shorter time period of growth

Compared to the direct seeding method, marcotting a tree branch shortens the time period from the first application of the method to gaining a ready-to-be-planted sapling. This is because the tree branch has leaves for photosynthesis, and the nutrient provided from photosynthesis will boost the growth of roots from the branch, while germination of seed takes longer time to grow.

b. Guaranteed same trait as parent plant

Seeds produced from mitosis (cell division) from male gamete and female gamete will shuffle the genes of the parent plant. To explain simply, saplings germinated from seeds will have random traits inherited from the parent plant, but not necessarily the same.

However, marcotting is essentially parts of the parent plant to form a new plant, so the saplings produced from marcotting are guaranteed to have the exact same trait as the parent plant. Therefore, if we perform marcotting on a parent plant that can harvest big and sweet fruit, we can get a new duplicated tree that produces the fruit of the same quality.

3. Disadvantages

The disadvantages of marcotting methods are stated as below:

a. High risk of infection

Since the same traits are inherited from the parent plant, if the parent plant is infected with disease, so will the saplings be infected with the same disease. If marcotting technique is applied on a large commercial scale, there will be a lot of loss.

b. No variant for evolution

Compared to the seed-germinated seedlings, saplings that are cultivated by marcotting will remain the same by any means, this also means that we could not change the traits of the plant, such as a variant of a shorter harvest period with bigger, sweeter fruit and immunity to disease.

4. Materials

Materials of marcotting are listed below:

1. Transparent Plastic Sheet
2. Peat Moss
3. Nylon String
4. Cutter
5. Rooting Powder



Figure 1: Materials of Marcotting

5. Techniques

This section records the steps of performing marcotting.

1. First, transparent plastic sheets were cut into squares of approximately 25cm x 25cm.
2. Next, a handful of peat moss (approximately 200g) were put in the middle of the plastic sheet. The peat moss was put under running tap water until damped. Excessive water was squeezed out from the medium.



Picture 2.1: A Handful of Peat Moss, Picture 2.2: Damped Peat Moss

3. A tree branch with healthy leaves is chosen. Note that the chosen tree branch must have a wooden stem instead of a young green stem, and has a diameter of at least 1cm. The

branch was prepared by removing unwanted leaves and small branches at the end of the stem to create a working space.

4. A cutter was used to slide carefully around the end of the chosen stem, sliding through the bark of the stem (brown, hard skin). Slide around the stem for a second time, at a distance of approximately 1cm away from the first cut. Then, the bark is peeled open by hand, also carefully removing any remaining bark.



Picture 3: Stem after bark is peeled away

5. Rooting powder was applied on the cut surface of the side near to the leaves of the branch.



Picture 4: Rooting Powder Application

6. The cut surface is covered with peat moss wrapped in plastic, both sides of the plastic are tightened using nylon string. Make sure the structure holds tightly onto the branch and will not shake, turn or fall off when touched.



Picture 5: Marcott

7. Leave the marcott for months and monitor the growth of the root. Once root has grown from the marcott, remove it and a young sapling is obtained and ready to be planted.

6. Theory

Marcotting is a technique to force root growth on branches. Peat moss is used as a medium because it is sterile to prevent bacteria inside the closed, damp marcott throughout the whole cultivation period of the sapling. When the bark is removed, we are removing the epidermis, cortex and phloem layer to cut off the nutrient from photosynthesis of the selected branch to be transported back to the main stem of the plant through the phloem. The remaining part contains xylem which is needed for transporting water from the root to leaf of the selected branches so it can carry on photosynthesis. Over time, the stem of the selected branch will swell because the nutrient cannot be transported further. Thus, with the aid of rooting powder, roots will be formed

in the condition of marcott and the selected branches will eventually detach from the original tree with roots and ready to be planted.

7. Reflection

Through this assignment, I learned about the correct way of cloning fruit trees by marcotting and the science behind it. I am more excited than I expected myself to be to be able to apply the knowledge of Biology about plant transport systems and fertilisation in a way I would never thought before. This process is still fresh in my mind as days pass by when I wrote this report and I believe that it will continue to be. I can't wait to share it with my family who is also interested in agriculture.



Picture 6: Selfie with my Marcott

8. Reference

1. National Parks Board. (2021). *Air-layering: A viable way of propagating woody plants*.

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